

Solving Equations with Squares (Homework)

Name
Date
Course

Solve.

① $x^2 = 9$

$x = \pm 3$

② $4y^2 = 16$

$y = \pm 2$

③ $b^2 + 5 = 54$

$b = \pm 7$

④ $\frac{x^2}{16} = 4$

$x = \pm 8$

⑤ $y^2 = 5$

$y = \pm \sqrt{5}$

⑥ $b^2 + 12 = 18$

$b = \pm \sqrt{6}$

⑦ $\frac{x^2}{5} + 2 = 10$

$x = \pm 2\sqrt{10}$

⑧ $100y^2 = 3$

$y = \pm \frac{\sqrt{3}}{10}$

⑨ $7b^2 = 4$

$b = \pm \frac{2\sqrt{7}}{7}$

⑩ $12x^2 + 9 = 14$

$x = \pm \frac{\sqrt{15}}{6}$

⑪ $6 + y^2 = 2$

No Solution

⑫ $(b-3)^2 + 1 = 10$

$b = 6$ or 0

⑬ $-2 = \frac{(x+2)^2}{4} - 6$

$x = 2$ or -6

⑭ $5(y-8)^2 + 2 = 17$

$y = \pm \sqrt{3} + 8$

⑮ $17 = 9(b-1)^2 + 18$

No Solution

⑯ $\frac{(x+7)^2}{-5} + 3 = 2$

$x = \pm \sqrt{5} - 7$

⑰ $5(y+10)^2 - 57 = 3$

$y = \pm 2\sqrt{3} - 10$

⑱ $17 = \frac{(b+9)^2}{3} + 11$

$b = \pm 3\sqrt{2} - 9$

⑲ $8 = 2(x-6)^2 - 1$

$x = \pm \frac{3\sqrt{2}}{2} + 6$

⑳ $-4(y+11)^2 + 6 = 3$

$y = \pm \frac{\sqrt{3}}{2} - 11$

㉑ $(3b+5)^2 - 2 = 34$

$b = \frac{1}{3}$ or $-\frac{11}{3}$

㉒ $17 = 1 + (\frac{x}{4} - 7)^2$

$x = 44$ or 12

㉓ $4(\frac{y}{2} + 3)^2 - 8 = 12$

$y = \pm 2\sqrt{5} - 6$

㉔ $-7 = -3 + 2(\frac{b}{6} - 1)^2$

No Solution

㉕ $\frac{(\frac{x}{3} + 2)^2}{5} - 8 = 2$

$x = \pm 15\sqrt{2} - 6$

$$\textcircled{26} -2(8b+1)^2+3=-19$$

$$b = \frac{\pm\sqrt{11}-1}{8}$$

$$\textcircled{27} 3 = 5\left(\frac{x}{4}-10\right)^2-4$$

$$x = \pm \frac{4\sqrt{35}}{5} + 40$$

$$\textcircled{28} -32\left(\frac{b}{7}-6\right)^2-1=-10$$

$$b = \pm \frac{21\sqrt{2}}{8} + 42$$

$$\textcircled{29} 19 = 8\left(\frac{y}{2}+7\right)^2-8$$

$$y = \pm \frac{3\sqrt{6}}{2} - 14$$